Batch:B2	**Roll No.: 16010124114**
Experiment No._____	
Grade: AA / AB / BB / BC / CC / CD /DD	
Signature of the Staff In-charge with date	

Title: Study, Implementation, and Analysis of Job Sequencing with Deadlines.

Objective: To learn the Greedy strategy of solving the problems for different types of problems

CO to be achieved:

CO 2 Describe various algorithm design strategies to solve different problems and analyse Complexity.

Books/ Journals/ Websites referred:

1. **Ellis horowitz, Sarataj Sahni, S.Rajsekaran," Fundamentals of computer algorithm", University Press**
2. **T.H.Cormen ,C.E.Leiserson,R.L.Rivest and C.Stein," Introduction to algorithms",2nd Edition ,MIT press/McGraw Hill,2001**
3. **http://lcm.csa.iisc.ernet.in/dsa/node184.htm**
4. **http://students.ceid.upatras.gr/~papagel/project/kruskal.htm**
5. **http://www.personal.kent.edu/~rmuhamma/Algorithms/MyAlgorithms/GraphAlgor/kruskalAlgor.html**
6. **http://lcm.csa.iisc.ernet.in/dsa/node183.html**
7. **http://students.ceid.upatras.gr/~papagel/project/prim.htm**
8. **http://www.cse.ust.hk/~dekai/271/notes/L07/L07.pdf**

Pre Lab/ Prior Concepts:

Data structures, Concepts of algorithm analysis

Historical Profile:The Job Sequencing with Deadlines problem is a classic optimization problem where the goal is to schedule jobs to maximize profit, ensuring each job is completed before its deadline. The algorithm follows a greedy approach.The Job Sequencing with Deadlines Problem is a foundational optimization problem in scheduling theory and computer science. Its historical development is closely tied to advancements in algorithm design, combinatorial optimization, and practical applications in operations research and industrial processes.

Origins and Early Development

Scheduling Problems in Industry (Mid-20th Century):The problem emerged during the industrial revolution and later in the mid-20th century with the rise of operational research.Scheduling tasks to maximize efficiency, minimize delays, or maximize profits became critical for industries like manufacturing and logistics.

Development of Mathematical Formulation:Researchers began formalizing scheduling problems, including constraints like deadlines and profit maximization, laying the groundwork for job sequencing problems.The focus shifted from heuristic or ad-hoc methods to formal algorithmic solutions.

New Concepts to be learned:
Application of algorithmic design strategy to any problem, Greedy method of problem solving Vs other methods of problem solving, optimality of the solution, knapsack problem and their applications

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Algorithm:

Input: A set of n jobs, where each job i has:

1. pi: Profit if the job is completed.
2. di: Deadline (maximum time slot by which it should be completed).

Output: A schedule of jobs such that:

1. Each job is completed within its deadline.
2. The profit is maximized.

Step 1: Sort Jobs by Profit. Sort all jobs in descending order of their profit pi. Jobs with higher profits are prioritized.

Step 2: Allocate Time Slots

1. Iterate through the sorted jobs.
2. For each job: Find the latest available time slot $t \leq di$ (the deadline of the job).If a time slot is available, schedule the job in that slot and update the slot as occupied.

Step 3: Output the Job Schedule.(Return the scheduled jobs and the total profit.)

Example:

~~Database (Hospital management)~~

Job scheduling with deadline.

Q.1

	J_1	J_2	J_3	J_4	J_5	J_6
Profit	15	10	12	20	8	5
Deadline	5	3	3	2	4	2

- Sort in descending order.
- Schedule the job (linear search to find particular slot)

J_6 (15)	J_4 (20)	J_3 (12)	J_5 (8)	J_1 (15)	
0	1	2	3	4	5

Profit = 70

Sorting $(O(n \log n))$ ~~+ max~~

Arrange $= O(n^2)$

Overall $= O(n^2)$.

Q.2

	J_1	J_2	J_3	J_4	J_5	~~J_6 J_7~~
Profit	100	19	27	25	15	
Deadline	2	1	2	1	3	

J_3 (27)	J_1 (100)	J_5 (15)
1	2	3

Sorted index	J_1	J_3	J_4	J_2	J_5
	100	27	25	19	15
	2	2	1	1	3

Profit = 142 //

Analysis of algorithm:

```
#include <bits/stdc++.h>
using namespace std;

int main() {
   vector<vector<int>> vc;
   int n;
   cout << "Enter the number of elements";
   cin >> n;

   for (int i = 0; i < n; i++) {
      int name;
      cout << "Enter the job name";
      cin >> name;

      int job;
      cout << "Enter the job profit :";
      cin >> job;

      int jobtime;
      cout << "Enter the job deadline:";
      cin >> jobtime;

      vc.push_back({job, jobtime, name});
   }

   sort(vc.begin(), vc.end(), [](vector<int>& a, vector<int>& b) {
      return a[0] > b[0];
   });

   int profit = 0;
   int times = 0;
   for (int i = 0; i < vc.size(); i++) {
      times = max(times, vc[i][1]);
   }

   vector<int> jobs(times + 1, -1);

   for (int i = 0; i < vc.size(); i++) {
      int start = vc[i][1];
      while (start > 0) {
         if (jobs[start] == -1) {
            jobs[start] = i;
            break;
         } else {
            start = start - 1;
         }
      }
   }
```

```
    for (int t = 1; t <= times; t++) {
        if (jobs[t] != -1) {
            int idx = jobs[t];
            cout << "J" << vc[idx][2] << " -> ";
            profit += vc[idx][0];
        }
    }

    cout << "The total profit is " << profit;
    return 0;
}
```

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<vector<int>> vc;
    int n;
    cout << "Enter the number of elements";
    cin >> n;

    for (int i = 0; i < n; i++) {
        int name;
        cout << "Enter the job name";
        cin >> name;

        int job;
        cout << "Enter the job profit :";
        cin >> job;

        int jobtime;
        cout << "Enter the job deadline:";
        cin >> jobtime;

        vc.push_back({job, jobtime, name});
```

```
Enter the number of elements6
Enter the job name1
Enter the job profit :15
Enter the job deadline:5
Enter the job name2
Enter the job profit :10
Enter the job deadline:3
Enter the job name3
Enter the job profit :12
Enter the job deadline:3
Enter the job name4
Enter the job profit :20
Enter the job deadline:2
Enter the job name5
Enter the job profit :8
Enter the job deadline:4
Enter the job name6
Enter the job profit :5
Enter the job deadline:2
J2 -> J4 -> J3 -> J5 -> J1 -> The total profit is 65

=== Code Execution Successful ===
```

main.cpp Share Run Output Clear

```
        vc.push_back({job, jobtime, name});
    }

    sort(vc.begin(), vc.end(), [](vector<int>& a, vector<int>& b) {
        return a[0] > b[0];
    });

    int profit = 0;
    int times = 0;
    for (int i = 0; i < vc.size(); i++) {
        times = max(times, vc[i][1]);
    }

    vector<int> jobs(times + 1, -1);

    for (int i = 0; i < vc.size(); i++) {
        int start = vc[i][1];
        while (start > 0) {
            if (jobs[start] == -1) {
                jobs[start] = i;
                break;
            } else {
```

```
Enter the number of elements6
Enter the job name1
Enter the job profit :15
Enter the job deadline:5
Enter the job name2
Enter the job profit :10
Enter the job deadline:3
Enter the job name3
Enter the job profit :12
Enter the job deadline:3
Enter the job name4
Enter the job profit :20
Enter the job deadline:2
Enter the job name5
Enter the job profit :8
Enter the job deadline:4
Enter the job name6
Enter the job profit :5
Enter the job deadline:2
J2 -> J4 -> J3 -> J5 -> J1 -> The total profit is 65

=== Code Execution Successful ===
```

main.cpp

```
        int start = vc[i][1];
        while (start > 0) {
            if (jobs[start] == -1) {
                jobs[start] = i;
                break;
            } else {
                start = start - 1;
            }
        }
    }

    for (int t = 1; t <= times; t++) {
        if (jobs[t] != -1) {
            int idx = jobs[t];
            cout << "J" << vc[idx][2] << " -> ";
            profit += vc[idx][0];
        }
    }

    cout << "The total profit is " << profit;
    return 0;
}
```

Output

```
Enter the number of elements6
Enter the job name1
Enter the job profit :15
Enter the job deadline:5
Enter the job name2
Enter the job profit :10
Enter the job deadline:3
Enter the job name3
Enter the job profit :12
Enter the job deadline:3
Enter the job name4
Enter the job profit :20
Enter the job deadline:2
Enter the job name5
Enter the job profit :8
Enter the job deadline:4
Enter the job name6
Enter the job profit :5
Enter the job deadline:2
J2 -> J4 -> J3 -> J5 -> J1 -> The total profit is 65

=== Code Execution Successful ===
```

Conclusion:

Successfully implemented the job scheduling with database.

Batch: **Roll No.:**

Experiment No._7_

Grade: AA / AB / BB / BC / CC / CD /DD